

XRPL Validator — Daily Engineering Report • 2026-08-13

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **4 commits** (xrpl-validator 4)

Engineering changes

- **fix(xrpl-ledger): stamount_signed_add ROUNDS its operands, not truncates** (6798100 , xrpl-validator) An STAmount operand is already canonical in rippled — it was rounded to 16 digits when it was produced — so reducing an over-precise operand by truncation drops exactly the digit that decides the result. norm16 truncates.
- **fix(xrpl-ledger): an INPUT-limited fill rounds the offer's output DOWN** (ca17418 , xrpl-validator) BookStep.cpp:653, verbatim, with rippled's own rationale above it: `ofrAmt = offer.limitIn(ofrAmt, inLmt, /roundUp / false);`
- **fix(xrpl-ledger): measure a hop's carry at full precision, not by differencing** (44c20d9 , xrpl-validator) The forward pass measured what a hop produced by differencing the sender's BALANCE. A balance holds 16 significant digits TOTAL, so whatever the sender already holds of the intermediate eats the low end of the addend — the hop is fine, the measurement is not.
- **fix(xrpl-ledger): reprice a rested remainder in BOTH directions** (afb133f , xrpl-validator) The one-way clamp on the reprice — take `derived` only when it moves the residual DOWN — was a workaround for #105930662 40FB322EC16C from when the reprice still used the RAW ratio. 070ca31 fixed the rate itself and that ledger is byte-exact now, so the guard had outlived its cause and did nothing except suppress the reprice wherever it needed to move the residual UP.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Oracle-Bound Batch Execution Guard** (payments) Add a `tfOracleGuard` flag to the outer Batch (XLS-56) transaction plus two SFields, `sfGuardOracleID` (a PriceOracle keylet) and `sfGuardPriceBounds` (min/
- **Credential-Based Trust Line Auto-Authorization** (compliance) Add a bounded SField `sfAutoAuthCredentials` (STVector256 of up to 8 {CredentialIssuer, CredentialType} pairs) to AccountRoot, settable via a new Account
- **Native Fee-on-Transfer for MPT Issuances** (tokens) Add `sfMPTTransferFee` (basis points, 0–50000, mirroring NFTokenTransferFee semantics) and `sfMPTFeeDestination` (optional AccountID, defaults to issuer)

Appendix — commits

Time	Repo	Commit	Subject
03:40	xrpl-validator	6798100	fix(xrpl-ledger): stamount_signed_add ROUNDS its operands, not truncates
12:05	xrpl-validator	ca17418	fix(xrpl-ledger): an INPUT-limited fill rounds the offer's output DOWN
13:37	xrpl-validator	44c20d9	fix(xrpl-ledger): measure a hop's carry at full precision, not by differencing
15:33	xrpl-validator	afb133f	fix(xrpl-ledger): reprice a rested remainder in BOTH directions